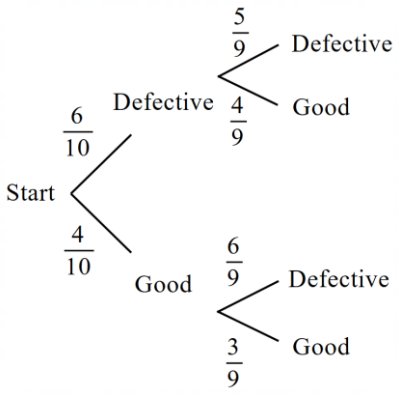
# 2025 VCE Mathematical Methods 2 (NHT) external assessment report

This report provides sample answers or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

## Section A – Multiple-choice questions

| Question | Correct answer | Comments |
|---|---|---|
| 1 | D | |
| 2 | B | |
| 3 | D | |
| 4 | B | |
| 5 | C | |
| 6 | D | |
| 7 | B | |
| 8 | C | |
| 9 | A | |
| 10 | A | $h(x)=g(f(x))=\left(e^{2x+3}\right)^2+2\left(e^{2x+3}\right)-3$<br>$h'(x)=2e^{2x+3}\times 2\left(e^{2x+3}\right)+4e^{2x+3}$<br>$=4e^{2x+3}\left(e^{2x+3}+1\right)$<br>$=4f(x)\left(f(x)+1\right)$ |
| 11 | C | |
| 12 | A | |
| 13 | D | |
| 14 | D | |
| 15 | C | |
| 16 | C | |

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| Question | Correct answer | Comments |
|---|---|---|
| **17** | A |  <br>Then the probability of at most one defective is<br>$\Pr(D \le 1) = 1 - \Pr(D = 2)$<br>$= 1 - \frac{6}{10} \times \frac{5}{9}$<br>$= \frac{2}{3}$<br>The probability of exactly one defective is<br>$\Pr(D = 1) = \frac{6}{10} \times \frac{4}{9} + \frac{4}{10} \times \frac{6}{9}$<br>$= \frac{8}{15}$<br>The required conditional probability is<br>$\Pr(D = 1 \mid D \le 1) = \frac{8}{15} \div \frac{2}{3}$<br>$= \frac{4}{5}$ |
| **18** | B | $f(x) = \log_e(x) - \frac{x}{3}$<br>$i = 1$<br>$c = \frac{4+6}{2} = 5$<br>$f(5) = -0.057...$<br>$f(4) \times f(5) = -0.003...$<br>$b = 5$<br>$i = 2$<br>$c = \frac{4+5}{2} = 4.5$<br>$f(4.5) = 0.004... < 0.01$<br>Print 4.5 |

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<table>
<tr><th>Question</th><th>Correct answer</th><th>Comments</th></tr>
<tr><td>19</td><td>A</td><td>
Consider the following table.<br>
For each cell, as long as $B^2 - 4C \geq 0$, put a tick.<br>
For example, when $B = 1$ and $C = 1$, $1 < 4 \times 1$, ×<br>
And when $B = 2$ and $C = 1$, $2^2 = 4 = 4$, ✓.
<table>
<tr><th>$C$ \ $B$</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr>
<tr><td>1</td><td>×</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>2</td><td>×</td><td>×</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>3</td><td>×</td><td>×</td><td>×</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>4</td><td>×</td><td>×</td><td>×</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>5</td><td>×</td><td>×</td><td>×</td><td>×</td><td>✓</td><td>✓</td></tr>
<tr><td>6</td><td>×</td><td>×</td><td>×</td><td>×</td><td>✓</td><td>✓</td></tr>
</table>
Each cell has a probability $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.<br>
Counting the number of ticks – there are 19.<br>
Hence, $\Pr\left(B^2 \geq 4C\right) = \frac{19}{36}$.
</td></tr>
<tr><td>20</td><td>A</td><td>
To have one unique solution, the gradients have to be different; for example, $y = 2x + 1$ and $y = 3x + 1$.<br>
For an infinite number of solutions, the gradients and $y$-intercepts have to be the same; for example, $y = 4x + 1$ and $2y = 8x + 2$.<br>
If these four linear equations are combined into a new system of four linear equations, there will be one solution, $(0, 1)$.<br>
Another example for an infinite number of solutions is $y = 4x + 2$ and $2y = 8x + 4$. When these are combined with $y = 2x + 1$ and $y = 3x + 1$, there will be no solution.
</td></tr>
</table>

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# Section B

## Question 1a

$g(x)=(x+1)^3(x-1)$

There is a stationary point of inflection at $x=-1$.

## Question 1b

The coordinates of the local minimum are $\left(\frac{1}{2},-\frac{27}{16}\right)$ or $(0.5,-1.6875)$.

## Question 1c

Average rate of change

$=\frac{g(2)-g(0)}{2-0}$

$=14$

## Question 1d

Solve $g'(x)=-2$

$x=-\frac{3}{2}$ or $x=0$

$y=-2x-\frac{43}{16}$, $y=-2x-1$

## Question 1e

1. Reflect in the $y$-axis

2. Translate by 1 unit right

or

1. Translate by 1 unit left

2. Reflect in the $y$-axis

## Question 1f

$a\times g\left(\frac{1}{2}\right)+b=0$

$b=\frac{27a}{16}$ or $b=1.6875a$

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# Question 1g.i

$k = 0$, $g_0(x) = x^4$, one solution

$k < 0$, $g_k(x) = \underbrace{\left(x^2 - k\right)}_{\text{no soln}}\underbrace{\left(x + k\right)^2}_{\text{one soln}}$

$k \leq 0$

# Question 1g.ii

$g_k'(x) = 2\left(x + k\right)\left(2x^2 + kx - k\right) = 0$

$x = -k,\ x = \dfrac{\pm\sqrt{k\left(k + 8\right)} - k}{4}$

Solve $-k = \dfrac{\pm\sqrt{k\left(k + 8\right)} - k}{4}$, $k = 0$

If $k \neq 0$

Solve $k\left(k + 8\right) < 0$, $-8 < k < 0$

$-8 < k \leq 0$

# Question 2a

$R^+$ or $(0, \infty)$

# Question 2b

$f(x) = 4e^{\log_e\left(x^3\right)}$

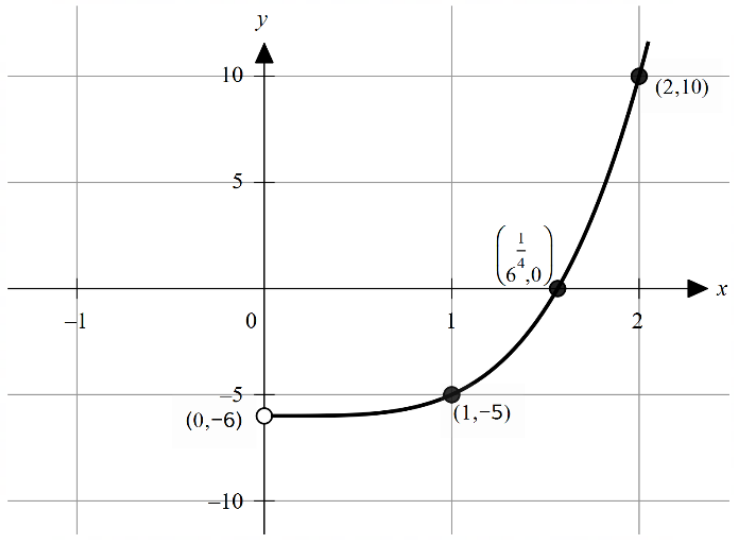
$= 4x^3$

# Question 2c

$A = \int_1^2 4x^3 dx = 15$

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# Question 2d.i




# Question 2d.ii

$$A=\int_{0.5}^{a} f(x)dx$$
$$=\left[g(x)\right]_{0.5}^{a}$$
$$=g(a)-g(0.5)$$
$$=g(a)-\left(-\frac{95}{16}\right)$$
$$=g(a)+\frac{95}{16}$$
$$b=\frac{95}{16} \text{ OR } 5.9375$$

or

$$A=\int_{0.5}^{a} f(x)dx$$
$$=\left[x^4\right]_{0.5}^{a}$$
$$=a^4-\frac{1}{16}$$
$$g(a)=a^4-6$$
$$b=6-\frac{1}{16}$$
$$b=\frac{95}{16} \text{ OR } 5.9375$$

# Question 2e.i

$84x^2-288x+216$ or $84\left(x-\frac{12}{7}\right)^2-\frac{216}{7}$

# Question 2e.ii

$x=\frac{12}{7}$ or $x=1.\overline{714285}$

$\left(\frac{12}{7},-\frac{972}{49}\right)$

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# Question 2e.iii

$p = \frac{12}{7}$

# Question 3a

$h(6) = 3.25$

# Question 3b

$\frac{1}{18-6}\int_{6}^{18} h(t)dt$

$= 2.98$

# Question 3c

$t_1 = 7.8035..., \quad t_2 = 7.9493...$
$t_3 = 8.27617... \quad t_4 = 10.6638...$

$(7.803... - 6) + (8.276... - 7.949...) + (18 - 10.663...)$

Total time $= 9.47$

# Question 3d

$h_{\max} = 5.20$

3.30 pm or $t = 15.5$ or 15 h 30 min or 930 min

# Question 3e

$$h(25k) = 2\sin\left(\frac{4\pi \times 25k}{25}\right) + \frac{1}{5}\sin\left(3\pi \times 25k\right) + 3$$
$$= \underbrace{2\sin\left(4k\pi\right)}_{=0} + \underbrace{\frac{1}{5}\sin\left(75k\pi\right)}_{=0} + 3 \quad \left(k \in Z^{+}\right)$$
$$= 3$$

# Question 3f

Period $= 50$

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# Question 4a

$$\mathrm{E}(X) = \int_{\frac{2}{3}}^{\infty} x \times \frac{8}{9x^4}\,dx$$

$$= 1 \text{ or } \$1000$$

# Question 4b.i

$$\Pr(X > 2) = \int_{2}^{\infty} f(x)\,dx$$

$$= \frac{1}{27} \text{ or } 0.\overline{037}$$

# Question 4b.ii

$$\Pr(X > 4 \mid X > 2) = \frac{\Pr(X > 4)}{\Pr(X > 2)}$$

$$= \frac{\int_{4}^{\infty} f(x)\,dx}{\int_{2}^{\infty} f(x)\,dx}$$

$$= \frac{\frac{1}{216}}{\frac{1}{27}}$$

$$= \frac{1}{8} \text{ OR } 0.125$$

# Question 4b.iii

$$L \sim \mathrm{Bi}\left(10, \frac{1}{27}\right)$$

$$\Pr(L \geq 2) = 0.051$$

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# Question 4b.iv

$$\Pr(\text{claim made} \cap X > 2) = \frac{1}{5} \times \frac{1}{27} = \frac{1}{135}$$

$$C \sim \text{Bi}\left(n = 2, p = \frac{1}{135}\right)$$

$$\Pr(\text{at least one claim with } X > 2)$$
$$= 1 - \Pr(C = 0)$$
$$= 1 - \left(\frac{134}{135}\right)^2$$
$$= 0.0148$$

or

$$\Pr(1 \text{ large and } 1 \text{ no claim}) = 0.2 \times 0.8 \times \frac{1}{27} \times 2 = 0.01185...$$

$$\Pr(1 \text{ large and } 1 \text{ not large}) = 0.2^2 \times \frac{1}{27} \times \frac{26}{27} \times 2 = 0.00285...$$

$$\Pr(2 \text{ large}) = 0.2^2 \times \frac{1}{27} \times \frac{1}{27} = 0.000054...$$

$$0.01185... + 0.00285... + 0.000054...$$
$$= 0.0148$$

or

$$1 - \Pr(\text{no claims large}) = 1 - 0.8^2 - 2(0.8)(0.2) \times \frac{26}{27} - 0.2^2 \times \left(\frac{26}{27}\right)^2$$

$$= 0.0148$$

# Question 4c.i

$$T_A \sim \text{N}(32, 7^2)$$

$$\Pr(T_A < 45) = 0.968$$

# Question 4c.ii

$$\Pr(T_A \leq k) = 0.995$$

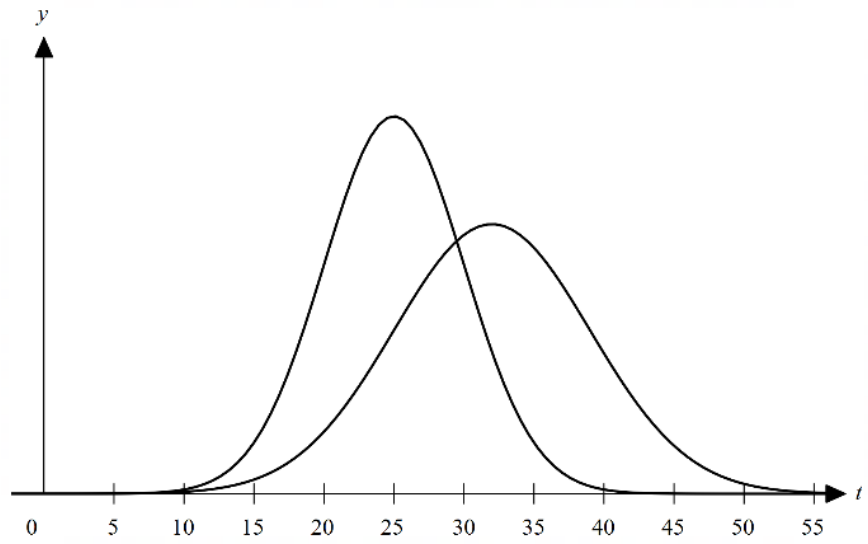
$$k = 50$$

# Question 4c.iii

(0.902, 0.998)

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# Question 4c.iv




# Question 5a.i

$2w+l=12,\ l=12-2w$

$$\begin{aligned}\text{Area} &= l\times w\\ &=(12-2w)\times w\\ &=12w-2w^2\end{aligned}$$

# Question 5a.ii

$12-4w=0,\ w=3$

or $A=-2(w-3)^2+18$

$A_{\max}=18$

# Question 5b.i

$3\log_e(k)=2\log_e(12-k)$

$k=4$

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# Question 5b.ii

Solve $g(x) = w$ for $x$

$x = e^{\frac{w}{3}}$, $x = 12 - e^{\frac{w}{2}}$

$l = 12 - e^{\frac{w}{2}} - e^{\frac{w}{3}}$

$A(w) = w \times l$

$= w\left(12 - e^{\frac{w}{2}} - e^{\frac{w}{3}}\right)$

# Question 5b.iii

$w = 2.49$, $A_{\max} = 15.52$

# Question 5b.iv

$l = \dfrac{15.5220...}{2.49083...} = 6.231666...$

Total length of timber $= 2 \times 2.49083... + 6.231666$

$= 11.2133... < 12$ m

Yes, David can build with less than 12 m timber

# Question 5c

$A_p(w) = w\left(12 - e^{\frac{w}{2}} - e^{\frac{w}{3}}\right) + \dfrac{1}{2}(g(4) - w)\left(12 - e^{\frac{w}{2}} - e^{\frac{w}{3}}\right)$

$2w + \sqrt{\left(4 - e^{\frac{w}{3}}\right)^2 + (g(4) - w)^2} + \sqrt{\left(12 - e^{\frac{w}{2}} - 4\right)^2 + (w - g(4))^2} \leq 12$

$2.63831... \leq w \leq 4.60057....$

Max value $= A_3(2.63831...) = 19.88$

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